

System Architecture — Internal Knowledge Assistant

1. Architectural Goals

- Accuracy through document-grounded responses
- Security of internal data
- Scalability and maintainability
- Clear separation of concerns

2. High-Level Components

User Interface

- Web-based interface for submitting questions
- Displays answers and citations

Backend API

- Orchestrates request flow
- Handles retrieval, prompt construction, and response delivery
- Exposes REST endpoints

Document Ingestion Service

- Parses documents (PDF, text)
- Chunks content
- Generates embeddings

Vector Database

- Stores document embeddings
- Enables semantic similarity search

LLM Service

- Generates natural-language responses
- Uses retrieved context only (RAG)

Monitoring & Logging

- Tracks request volume
- Measures latency and errors
- Supports future observability tooling

3. Data Flow

1. Admin uploads documents
2. Documents are chunked and embedded
3. Embeddings stored in vector database
4. User submits question
5. System retrieves top-k relevant chunks
6. Prompt constructed with context

7. LLM generates grounded response
8. Answer and citations returned to user

4. Deployment Model (Target)

Layer	Technology
UI	Static hosting
API	FastAPI on EC2 / Container
Storage	S3
Vector DB	FAISS (MVP), OpenSearch (future)
AI	OpenAI / Azure OpenAI
Monitoring	CloudWatch

5. Key Design Decisions

- Retrieval-Augmented Generation instead of fine-tuning
- Managed cloud services to reduce operational overhead
- Modular architecture to enable future expansion

6. Future Enhancements

- Role-based access control
- Authentication & authorization
- Feedback-driven answer improvement
- Cost optimization and caching
- Multi-region deployment